Randomized and low-rank approximation

Polynomial trace-error factor after exactly the target rank of pivots

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because removing exponential loss without oversampling requires sharper adaptive-pivot analysis; community impact is rank-efficient PSD approximation.
Topic: randomized factorization; approximation guarantees

Resolution — 2026-09-11

Negative resolution by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Theorem 1 and Corollary 3 disprove the existence of constants C, p giving the displayed polynomial bound after exactly r pivots. For every fixed $r \geq 1$, real entrywise-positive positive-definite matrices of order $r+1$ approach the sharp expected trace-error ratio 2^r as a parameter tends to zero. Choose r first and then the parameter; no limit uniform in r is needed. The result does not address oversampling (RA-01).

Complete manuscript · TeX · Independent complete-source PASS review · Authorship, exact checks and provenance.

Verification is independent agent review, not external human peer review or formal certification. AI assistance is disclosed; no priority claim is made. The original statement and audits remain below; ratings are historical.

Context and notation

For a Hermitian positive-semidefinite $A \in \mathbb{C}^{n\times n}$, define the exact-arithmetic RPCholesky residuals by $R_0 = A$. Conditional on $R_t \neq 0$, select j with probability $(R_t)_{jj}/\operatorname{tr}(R_t)$ and set

$$R_{t+1} = R_t - \frac{R_t(:,j)R_t(j,:)}{(R_t)_{jj}}.$$

If $R_t = 0$, keep all subsequent residuals zero. Eigenvalues are ordered $\lambda_1(A) \geq \cdots \geq \lambda_n(A) \geq 0$, and $\tau_r(A) = \sum_{j>r} \lambda_j(A)$. This is the pivot rule in Chen, Epperly, Tropp, and Webber, *Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations*, Algorithm 1; their Lemma 5.5 gives the current comparison $\mathbb{E}\operatorname{tr}(R_r) \leq 2^r \tau_r(A)$.

Problem statement

Do constants $C > 0$ and $p \geq 0$ exist such that, for every $n \geq 1$, every Hermitian positive-semidefinite $A \in \mathbb{C}^{n\times n}$, and every integer $1 \leq r \leq n$,

$$\mathbb{E}\operatorname{tr}(R_r) \leq C r^p \tau_r(A)?$$

Both constants must be independent of n, r, A. Exactly r RPCholesky steps are permitted, with the zero-residual convention above. This asks for a polynomial approximation factor without oversampling; RA-01 instead allows additional pivots to obtain relative error near one.

References

Epperly, *Make the Most of What You Have*, §11.1, Conjecture 11.2, p. 182; Chen et al., RPCholesky, Lemma 5.5, p. 1020. Gilles and Wilber, *Low-Rank Approximation by Randomly Pivoted LU*, §3.1, paragraph following Theorem 3, explicitly reiterate this conjecture.

Source normalization

The dissertation calls $A^{(r)}$ a residual but prints $\mathrm{tr}(A - A^{(r)})$, without expectation. We use its cited Lemma 5.5 to correct both notation defects. The later paper confirms the intended polynomial improvement over that lemma's 2^r factor.

Status check

Searches for *RPCholesky r-step polynomial*, *RPCholesky conjecture polynomial*, and the cited papers found no resolution. Epperly's August 2026 oversampling theorem does not supply a polynomial factor at exactly r steps. Its Theorem 1.2 is noninformative at $k = r$.

Audit — 2026-09-10

Rechecked Gilles–Wilber's discussion after Theorem 3, which reiterates the polynomial-factor conjecture. Exactly-r-step and later RPCholesky searches found no resolution. The August oversampling theorem does not establish the displayed factor after exactly r pivots.